

Problem Setting

Non-convex:
 $\min_{x \in \mathcal{X}} F_1(x) \quad \text{s.t. } F_2(x) \leq 0$
 $\mathcal{X} \subset \mathbb{R}^d$ — convex

Convex (unknown):
 $\min_{u \in \mathcal{U}} H_1(u) \quad \text{s.t. } H_2(u) \leq 0$
 $\mathcal{U} = c(\mathcal{X})$ — convex

(HC) Hidden Convexity [1]

- $\min_{u \in \mathcal{U}} H_1(u)$ s.t. $H_2(u) \leq 0$ is convex and admits a solution.
- Exists an invertible $c : \mathcal{X} \rightarrow \mathcal{U}$ with $\mu_c > 0$ such that

$$(HC) \quad \|c(x) - c(y)\| \geq \mu_c \|x - y\| \quad \forall x, y \in \mathcal{X}.$$

Oracle access: (Sub-)gradients $g_i \in \partial F_i(x)$, function evaluations $F_2(x)$.

Main contribution.

First solution methods for hidden convex constrained optimization with provable global convergence.

Summary of Total (Sub-)Gradient Complexities

Assumptions	Setting	Method	Complexity
(WC), (BG), (HC)	$F_2(\cdot) \equiv 0$	Sub-Grad. Method	$\tilde{\mathcal{O}}(\varepsilon^{-6})$ [1]
	$F_2(\cdot) \not\equiv 0$	S-IPPM + SwSG	$\tilde{\mathcal{O}}(\varepsilon^{-3})$
(Sm), (HC)	$F_2(\cdot) \equiv 0$	Gradient Descent (GD)	$\tilde{\mathcal{O}}(\varepsilon^{-1})$ [2]
	$F_2(\cdot) \not\equiv 0$	S-IPPM + ACGD	$\tilde{\mathcal{O}}(\varepsilon^{-2})$
	$F_2(\cdot) \not\equiv 0$	S-StarBL (F^* is known)	$\tilde{\mathcal{O}}(\varepsilon^{-1})$

“S-IPPM” is Shifted Proximal Point, “SwSG” is Switching Subgradient, “ACGD” refers to Accelerated Constrained GD, and “S-StarBL” is Shifted Star Bundle-level.

Motivating Application

Convex Constrained MDP [2]. State-action occupancy measure:

$$x := \pi \quad \mapsto \quad u := \lambda^\pi(s, a) := \sum_{h=0}^{+\infty} \gamma^h \mathbb{P}_{\rho, \pi}(s_h = s, a_h = a) \quad (\text{SOM})$$

Policy Optimization — typically non-convex in $\pi \in \Pi \subset \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|}$:

$$\begin{aligned} \min_{\pi \in \Pi} \quad & F_1(\pi) := H_1(\lambda^\pi), \\ \text{s.t.} \quad & F_2(\pi) := H_2(\lambda^\pi) \leq 0, \end{aligned} \quad (\text{CCMDP})$$

but convex in $u = \lambda^\pi$ since $H_1, H_2 : \mathcal{U} \rightarrow \mathbb{R}$ are convex utilities.

- *Standard Constrained RL*, $H_1(\lambda^\pi) = r^\top \lambda^\pi$, $H_2(\lambda^\pi) = c^\top \lambda^\pi$ are linear.
- *Safe Exploration*, $H_1(\lambda^\pi)$ – negative entropy, $H_2(\lambda^\pi) = c^\top \lambda^\pi$ linear.
- *Imitation learning*, $H_1(\lambda^\pi)$ or $H_2(\lambda^\pi)$ is a KL divergence.

Assumptions

(WC) Functions F_1 and F_2 are ρ -**weakly convex**, i.e., $F_i + \frac{\rho}{2} \|\cdot\|^2$ are convex.

(BG) Bounded subgradients: $\|g_i\| \leq G$ for all $x \in \mathcal{X}$, $g_i \in \partial F_i(x)$, $i = 1, 2$.

(Sm) F_i are differentiable, and $\|\nabla F_i(x) - \nabla F_i(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathcal{X}$.

Approach I: Shifted Proximal Point

At each iteration, solve the **shifted** strongly convex subproblem:

$$\begin{aligned} x^{(k+1)} &\approx \arg \min_{x \in \mathcal{X}} \varphi_1^{(k)}(x) := F_1(x) + \rho \|x - x^{(k)}\|^2, & (\text{S-IPPM}) \\ \text{s.t. } \varphi_2^{(k)}(x) &:= F_2(x) + \rho \|x - x^{(k)}\|^2 - \tau \leq 0. \end{aligned}$$

Idea: Prove that x_α is a Slater point of (S-IPPM); use feasible algo:

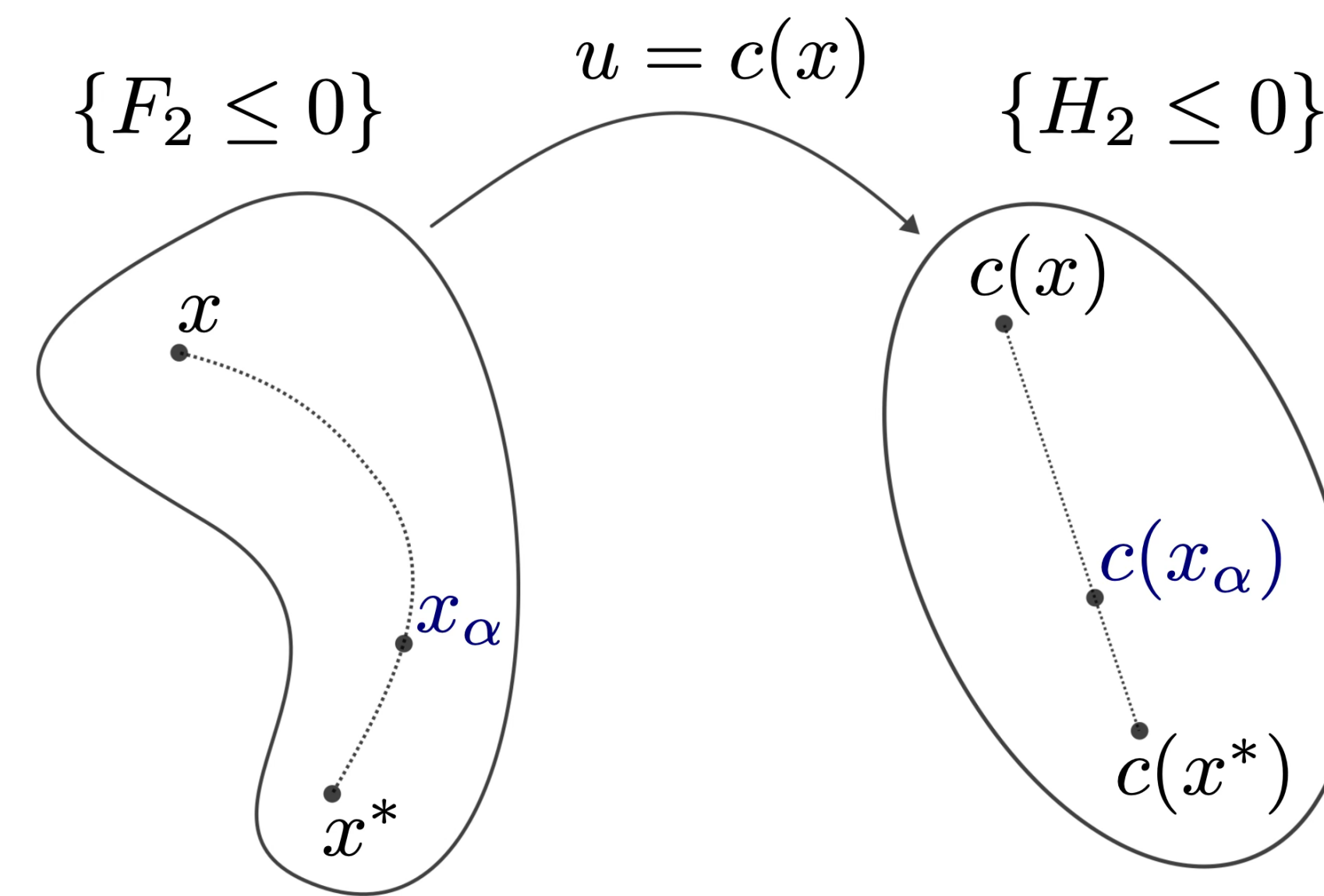
- (S-IPPM+SwSG) for weakly convex [3],
- (S-IPPM+ACGD) for smooth [4].

Approach II: Shifted Bundle-level

$$\begin{aligned} x_{t+1} &= \arg \min_{x \in \mathcal{X}} \|x - x_t\|^2, & (\text{S-StarBL}) \\ \text{s.t. } F_1(x_t) &+ \langle \nabla F_1(x_t), x_t - x \rangle \leq F_1^* + \tau_1, \\ F_2(x_t) &+ \langle \nabla F_2(x_t), x_t - x \rangle \leq \tau_2. \end{aligned}$$

Idea: Bundle-level methods [5] fail in non-convex case $\Rightarrow$ add shifts τ_1, τ_2 .

Note: If F_1^* is unknown, then estimate it $\Rightarrow$ S-BL+AdaLS.



Proof visualization: (left) non-convex space, (right) convex space.

References:

- [1] I. Fatkhullin, N. He, Y. Hu. Stochastic optimization under hidden convexity. SIAM Journal on Optimization, 2025.
- [2] J. Zhang, C. Ni, C. Szepesvari, M. Wang. *On the convergence and sample efficiency of variance-reduced policy gradient method*. NeurIPS 2021.
- [3] B. Polyak. A general method for solving extremum problems. Soviet Math. D., 1967.
- [4] Z. Zhang and G. Lan. Solving convex smooth function constrained optimization is almost as easy as unconstrained optimization. arXiv: 2210.05807, 2022.
- [5] C. Lemaréchal, A. Nemirovskii, and Y. Nesterov. New variants of bundle methods. Mathematical Programming, 1995.

Theoretical Analysis

S-IPPM needs only $\mathcal{O}(1/\varepsilon \log 1/\varepsilon)$ outer iterations

Thm. Under (HC), (WC) and $\text{diam}(\mathcal{U}) \leq \mathcal{D}_{\mathcal{U}}$, $\Delta_0 := F_1(x^0) - F_1^*$,

$$N \geq \frac{3\rho \mathcal{D}_{\mathcal{U}}^2}{\mu_c^2 \varepsilon} \log\left(\frac{3\Delta_0}{\varepsilon}\right) \implies F_1(x^N) - F_1^* \leq \varepsilon, \quad F_2(x^N) \leq \varepsilon.$$

If each (S-IPPM) solved to $\mathcal{O}(\varepsilon^2)$ -accuracy, then $T_{\text{total}} = N \cdot T_{\text{inner}}(\varepsilon)$:

$$\begin{aligned} \text{SwSG (BG)} & & \text{ACGD (Sm)} \\ T_{\text{inner}}(\varepsilon) &= \tilde{\mathcal{O}}(\varepsilon^{-2}) & T_{\text{inner}}(\varepsilon) = \tilde{\mathcal{O}}(\varepsilon^{-1}) \\ T_{\text{total}} &= \tilde{\mathcal{O}}(\varepsilon^{-3}) & T_{\text{total}} = \tilde{\mathcal{O}}(\varepsilon^{-2}) \end{aligned}$$

S-StarBL needs only $\mathcal{O}(1/\varepsilon \log 1/\varepsilon)$ total iterations

Thm. Under (HC), (WC), (Sm) and $\text{diam}(\mathcal{U}) \leq \mathcal{D}_{\mathcal{U}}$,

$$T \geq \frac{(\rho + L) \mathcal{D}_{\mathcal{U}}^2}{\mu_c^2 \varepsilon} \log\left(\frac{2v(x^{(0)}, F_1^*)}{\varepsilon}\right) \implies F_1(x_T) - F_1^* \leq \varepsilon, \quad F_2(x_T) \leq \varepsilon,$$

where $v(x^{(0)}, F_1^*) := \max\{F_1(x^{(0)}) - F_1^*, F_2(x^{(0)})\}$.

Numerical Illustration

Constrained Geometric Programming: $F_i(x) = \sum_{k_i=1}^{K_i} b_{k_i} x_1^{a_{1,k_i}} \cdots x_d^{a_{d,k_i}}$.

